Multimodal Fake News & Image Tampering Diagnostic Report

Generated on 2026-08-02 16:42:02 IST | Project Forensic Audit
Analysis performed at 2026-08-02 16:41:35 IST

1. Executive Summary & Verdict

Final Classification Verdict	**FAKE**
Overall Multimodal Risk Score	0.5465 (0.0=Real, 1.0=Fake)
Prediction Confidence	57.34%
Processing Time	1.58 seconds

This submission was classified as likely fabricated or misleading, with limited confidence (57.3%). The article's text contributed most to this verdict (64% of the fused score), primarily because sensationalist/conspiratorial language detected: anonymous. Additional supporting evidence: no attribution to an official source, study, or named authority was found in the text.

2. Evidence Summary

• Sensationalist/conspiratorial language detected: anonymous.
• No attribution to an official source, study, or named authority was found in the text.
• Text model attribution flagged these tokens as most fake-indicative: hidden, anonymous, claim.
• Error Level Analysis found no strong splicing/re-compression signal (anomaly score 0.116).
• Noise-inconsistency analysis found unusually uneven noise levels across the image, a pattern sometimes associated with compositing.
• No EXIF metadata was found in the image file.
• Simulated propagation graph shows an elevated proportion of suspicious/bot-like spreader accounts (22%).

3. Model Results & Fusion Weights

Modality	Model Engine	Risk Score	Weight	Relative Impact
Text Module	TF-IDF / BERT Transformer	0.8690	0.40	63.6%
Image Module	ResNet50 / ELA / pHash	0.4240	0.35	27.2%
Graph Module	PyG Graph Attention (GAT)	0.2018	0.25	9.2%

4. Analyzed Article Content

"A newly discovered herbal drink made from rare Himalayan plants has been proven to eliminate diabetes, cancer, heart disease, and Alzheimer's within just seven days. Anonymous experts claim that pharmaceutical companies have hidden this discovery for years to protect their profits. The drink is said to work instantly without any medical treatment or side effects. image of AI-generated herbal bottl..."

5. Text Explanation — Top Contributing Tokens

SHAP/coefficient attribution identified the following tokens as most influential in the text-model verdict: **hidden, anonymous, claim, years, experts, companies, effects, pharmaceutical**.

6. Image Explanation

Error Level Analysis showed no strong splicing signal (anomaly score 0.1158). Perceptual hash: `861939a00c602307`. A GradCAM attention heatmap localizing the regions most influential to the ResNet-50 verdict was generated for this image (see dashboard for the visual overlay).

Forensic observations: no EXIF metadata was found (common after re-saving, screenshotting, or social-media re-upload, but also consistent with metadata having been deliberately stripped); noise levels vary unusually across regions of the image (coefficient of variation 0.89), a pattern sometimes associated with compositing content from a different source.

7. Propagation Analysis Summary

Simulated propagation graph: 45 nodes, 118 cascade edges, 5 detected communities, diffusion depth 5, 22% suspicious/bot-like accounts. Graph Attention Network propagation risk score: 0.2018.

8. Fusion Analysis

The final verdict is a weighted linear combination of the unimodal risk scores (see Section 3), calibrated via Platt scaling fit on the project's labeled demo corpus. Any modality not submitted for this analysis defaults to a neutral 0.5 score rather than being fabricated. The modality with the largest 'Relative Impact' in the table above contributed most to the final classification.

9. Final Decision & Recommendations

This content shows characteristics consistent with misinformation. Before sharing or acting on it: verify the claim against at least one established fact-checking organization or primary source, check whether the accompanying image appears elsewhere in a different context (reverse image search), and treat unattributed claims and emotionally-charged language as a caution signal.